

Sabanci University
Faculty of Engineering and Natural Sciences
CS 419 Digital Image and Video Analysis
2025-2026 Fall

Course Coordinator	Lectures	Office Hours
Erchan Aptoula	Wednesday 08.40 - 09.30 FENS L045	By appointment
	Thursday 08.40 - 10.30 FASS G062	
	Lab Sessions	
	Monday 14.40-16.30 FENS L045	
Teaching assistant		
Berk Kivilcim		

Tentative schedule

Week	Subject	
1	Introduction and basics	
2	Point processing	Project topics announced, Lab
3	Binary image analysis	Homework 1 posted, project proposal and group formation deadline, Lab
4	Spatial image enhancement	Homework 1 deadline, Lab
5	Image enhancement in the frequency domain	Homework 2 posted, Lab
6	Non-linear filtering	Homework 2 deadline, Lab
7	Color image processing	Lab
8	Midterm exam	Midterm exam
9	Image segmentation	Progress report deadline, Lab
10	Global content description: color	Homework 3 posted, Lab
11	Global content description: shape	Homework 3 deadline, Lab
12	Global content description: texture	Lab
13	Local content description and bag of words	Presentations (either in class or by recording depending on crowd)
14	Video processing and tracking	Final report deadline

Catalog information: This course provides a comprehensive introduction into digital image & video processing and analysis. Major topics include image acquisition, linear and non-linear filtering, color, content description and video analysis. Students will learn the basic concepts of image and video processing as well as acquire hands-on experience in solving real-life visual analysis problems.

Textbooks

- Gonzales and Woods, Digital Image Processing, 4th Ed, 2017
- W. Pratt, Digital Image Processing, 4th Ed.
- P. Soille, Morphological Image Analysis, 2004
- R. Szeliski, Computer Vision and Applications, 2010

Prerequisites: Python programming, calculus, linear algebra, elementary probability and statistics.

Grading

Take-home assignments (30%): probably 3, involving programming (only in Python, Java or C/C++), mathematical tasks and design questions.

Group project (35%): proposal (5%), progress (8%) and final report (10%), presentation (9%) and peer grade (3%)

Midterm exam (35%): in-class, closed-book exam, with mostly design questions (no computers). There will only be one make-up exam, and only for students with official excuses (medical, etc).

Attendance: will be collected regularly from each lecture and laboratory session. If you miss 1/3 of the signature collections from the lectures or 1/3 of the signature collections from the lab sessions, your grade will be NA. If you miss the midterm exam (with no official excuse) your grade will be NA. The TA will be responsible for all matters regarding attendance (e.g. "I was at the restroom when the signatures were collected, can I sign now?", etc).

Laboratory sessions: the lab sessions will take place every week or every other week depending on the accumulation of topics. The lab sessions will not be graded.

Communication: All communication will run through SUCourse+ (questions, announcements, etc.).

Honor code: All tasks are individual unless stated otherwise. You can of course collaborate with other students, discuss with them. However, each student must submit their own work, which must be done only by themselves. If another student has helped or contributed in any way, this must be reported explicitly and in detail. Do not share any course content publicly/online, without explicit permission by the course coordinator. Plagiarism checks will be conducted regularly and all cases of confirmed cheating (all involved parties) will be reported for disciplinary action. Using AI tools is prohibited unless stated otherwise; If allowed, and you do use AI tools, you must provide links to the chat logs.

Late day policy: each student will have 48 "free" late hours. You can use them for any take-home assignment and only for take-home assignments. Late submission durations will be rounded to the next nearest hour; e.g. submitting 10 minutes late counts as 1 hour late. Past those 48 hours, submissions will not be graded and will not be accepted.

Group project: the students will form groups (of n individuals where n will depend on the number of students and will be specified after the add-drop week). Each group will undertake one project, the topic of which will be chosen from a predefined list of topics provided by the instructor. This project involves the following deliverables:

- a) a proposal document, explaining the project's scope, challenge, the dataset(s) to be used, and the methods to be explored, as well the performance and success metrics.
- b) a progress report (towards the middle of the semester – no page limit) explaining the so far progress, and the upcoming stages
- c) a final report at the last week of teaching. The report will be prepared in LaTeX and will be at least 6 pages in double column format (IEEE conference template). The final report should explain: the problem you are trying to solve (context, existing solutions if any, challenge), your approach (including diagrams and math), your results (all experiments you have conducted, their discussion, qualitative and quantitative results, ablation study) and the task distribution among group members.
- c) a presentation, as a pre-recorded video or in class if the number of presentations is low enough.

The final project grade will involve a peer grade as well.

Auditing: only with permission from the instructor.

Office hours: only via appointment (send an email).

Regrading policy: submit a regrade request to the TA; however regrading will take place for the entire graded unit. All regrading requests/objections must be made at most 3 days after the grade announcements.

Remarks:

- 1) The instructor has the right to perform an oral exam for students suspected of cheating.
- 2) Students are responsible for the material covered in class, or decisions/announcements/changes made in class.
- 3) Students are responsible for checking their emails (official Sabanci University accounts) and SUCourse+ for any course related announcement.
- 4) There will be no additional tasks, assignments or projects of any type for extra credit.
- 5) Reference letters will be provided only to students that have succeeded this course with a letter grade of "A".